COMPANION TO

Capability Matters: A Casebook

LENS Companion

CONCENTRATION · CROSSWALKS · REFERENCES

Learning Engineering for Next-Generation Systems · v2.1 (June 2026)

LDT · SCHOOL OF EDUCATION · JOHNS HOPKINS UNIVERSITY

SECTION

How to use this companion

The LENS companion is a single, white-paper reference that pairs the program's concentration documentation with the casebook's crosswalks and indexes. It is built from the same source as **Capability Matters: A Casebook** but renders no case bodies — the casebook is the place to read the cases themselves; the companion is the place to find them.

The companion is in five parts:

- **I.** The Five LENS Competencies — formal names, taglines, main objectives, and the v2.1 subobjective set (1.1 – 5.6). The organising layer over the formal CLOs.
- **II.** The CLOs and the Course Mapping — the five concentration learning objectives, the LDT / LENS course descriptions, and the course-to-CLO coverage matrix.
- **III.** Crosswalks — the casebook's induced framework mapped to the canonical five competencies, and the LECF crosswalk to the IEEE ICICLE community work.
- **IV.** Casebook indexes — every case by primary domain, every case by LENS course, and the CLO + course coverage appendix.
- **V.** References by case — the appendix the casebook ships with, so a reader can find every source the volume draws on.

> **On versioning.** This companion documents the **v2.1** program state (June 2026), in which D3 is renamed **Human-System Collaboration** and moved from position 5; T&E moves to D4; Sociotechnical Constraints moves to D5. The order now reads as the flywheel: see the system → build → integrate humans → measure → deploy. Prior versions are preserved in the program docs' change logs (`lens_program/1_LENS_Five_Competencies.md`).

SECTION

Why a companion — the thesis in brief

Learning engineering is a toolbox, not a single tool. At its centre is a dialogue between two traditions — the engineering disciplines that design, build, and sustain complex systems, and the learning sciences and education that study how people come to know and do. Around that dialogue gather many other fields: cognitive and human-factors engineering, measurement, implementation science, design, data and computation, and the operational domains themselves. The work is plural by design.

Capability is a system parameter.

Every complex system depends on what people can do inside it. That dependency is measurable, designable, and too important to leave to chance.

01. Systems don't fail because the technology breaks. They fail because someone couldn't do what the system required.

02. We engineer every parameter of a system except the people.

03. Capability is not a soft problem. It is a systems engineering problem.

04. Decision-grade evidence. Not opinions about training.

05. Capability without agency is automation. The goal is operators who can perform, adapt, and lead — not comply.

06. Sometimes the constraints make the goal unreachable. Saying so early is a result, not a failure.

Capability lives at the **interface** between what a system requires of its operators and the impact the system has to deliver. When that interface is wrong, the gap may originate in human development, in system design, in organisational performance, or in the interaction among them. Distinguishing which is itself a measurement and engineering problem; the casebook calls it **gap attribution**, and treats it as the diagnostic move that the rest of the discipline depends on.

The premise of the Johns Hopkins School of Education's Learning Design & Technology program — and of the Learning Engineering for Next-Generation Systems specialisation that has grown out of it — is that the capability parameter is engineerable. Not by accident. Not by declaration. By the same kind of evidence-grounded, methods-grounded, cross-domain discipline that built modern reliability engineering, modern epidemiology, and modern systems safety. The discipline does not yet exist at the scale the evidence demands. LDT and LENS are organised to help build it.

> **The casebook's method, in one sentence.** Each case is an **unpacking**: take a real outcome, recover the capability question behind it, and attribute the gap to human development, system design, or organisational performance. The discipline of unpacking is what LENS teaches; the casebook is its primary instrument. The framework that follows is the analytic lens the unpacking method uses — five competencies, each a way of asking what the work the system requires actually is, and whether the people inside it can do it.

SECTION

The Five LENS Competencies — v2.1

Organising layer over the formal CLOs. Formal names carry program documentation; taglines carry recruiting and the site. Subobjective numbers (1.1, 3.2, …) are stable tags for case studies, course modules, and capstone rubrics. The CLO language in the program documentation remains the language of record.

1 Systems Analysis

SYSTEMS THINKING AND ANALYSIS (CLO-1)

see the whole system

Main objective. Trace how human capability and system performance depend on each other, so interventions target the real problem.

1.1 Decompose system performance requirements into measurable human capability requirements: define what the system demands of its operators and the impact it must deliver.

1.2 Model learning environments and their host operational systems as interacting systems, with defined components, interfaces, and feedback loops.

1.3 Apply systems engineering lifecycle models to scope, sequence, and evaluate capability interventions within and across disparate systems.

1.4 Analyze and communicate human-system interdependencies to identify capability gaps and predict operational impact at scale.

1.5 **Governance-objection diagnostic.** Distinguish a governance objection that good design can dissolve from one that correctly signals the system should not deploy. [v2]

2 Iterative Development

LEARNING ENGINEERING DESIGN AND IMPLEMENTATION (CLO-2)

build, test, refine

Main objective. Design capability interventions through iterative engineering cycles that survive contact with real operational environments.

2.1 Design interventions that integrate learning sciences principles with measurable outcomes and system design constraints.

2.2 Run the iteration cycle: design, instrument, evaluate, refine, and revise the problem framing as evidence accumulates.

2.3 Evaluate evidence-based design strategies for transfer to high-consequence settings at operational speed and scale.

2.4 Construct implementation plans that address adoption, sustainment, and lifecycle integration across organizational and system boundaries.

2.5 **Narrate and defend the design iteration in first person.** Design competence includes not only producing an outcome but rendering the iteration legible. [v2]

3 Human-System Collaboration

HUMAN-SYSTEM COLLABORATION AND ADAPTIVE SYSTEMS (CLO-3)

work together

Main objective. Design human-system partnerships — including but not limited to human-AI — that make people more capable while preserving human agency and the recoverability of the system.

3.1 Design role architectures, interface and alert systems, mode/state transparency, authority gradients, and recoverability mechanisms that engineer collaborative capability at the human-system boundary.

3.2 Evaluate the measured impact of system-mediated work on human performance: the gains the system enables and the risks it introduces — automation bias, cognitive offloading, skill atrophy.

3.3 **Delegation with revocation.** Design the human oversight layer for delegated work; specify in advance the disconfirming evidence that would revoke the delegation. [v2]

3.4 **Collaboration measurement.** Measure the capability of a team or collaboration as a unit of analysis, distinct from any individual operator or any individual system component. [v2]

3.5 Specify learning and capability requirements for systems not yet fielded, working from design artifacts rather than operational experience.

4 Test and Evaluation

DATA, MEASUREMENT, AND EVALUATION (CLO-4)

show what works

Main objective. Produce decision-grade evidence linking learning to operational impact — where decision-grade is a sufficiency judgment under irreducible uncertainty — and tell a learning gap from a system gap.

4.1 Design ethical instrumentation strategies that measure capability at the speed and scale operations require.

4.2 **Gap attribution.** Diagnose the gap: differentiate capability shortfalls rooted in human development from those rooted in system design or organizational performance.

4.3 Construct decision-grade evidence artifacts under irreducible uncertainty: state what is known, what is assumed, and what would change the decision.

4.4 **Judgment under inadequate evidence.** Justify a consequential decision on incomplete and contested evidence; document the basis; name what would change it. [v2]

4.5 Communicate evidence honestly to technical and non-technical stakeholders, including uncertainty and limits of inference.

4.6 **Fairness beyond omission.** Demonstrate that omitting a protected attribute does not establish fairness; analyze competing fairness definitions using demographic-stratified outcome evidence. [v2]

5 Navigating Sociotechnical Constraints

CONTEXT AND DOMAIN FLUENCY (CLO-5)

make it work in the real world

Main objective. Integrate capability interventions into the sociotechnical systems they must live in: the regulatory, organizational, cultural, and technical realities that determine whether good designs survive.

5.1 Analyze the regulatory, organizational, cultural, and technical constraints that shape what can be built, deployed, and sustained in a given context.

5.2 Apply human systems integration frameworks to fit capability interventions to operational environments.

5.3 Leverage prior domain expertise, and respect for others', to elicit and validate specialist knowledge that surfaces constraints early.

5.4 Translate constraints and stakeholder requirements across disciplinary and institutional boundaries into coherent design specifications.

5.5 Anticipate adoption and sustainment barriers, and design interventions that survive them.

5.6 **Cross-regime / platform-dependency governance.** Where capability is deployed on a platform governed by a different regime than the one operating it, design the governance seam as an explicit deliverable. [v2]

See the whole system. Build, test, refine. Work together. Show what works. Make it work in the real world.

SECTION

CLOs and Course Mapping — v2.1

The LDT M.Ed. is a 10-course, 30-credit, online/asynchronous master's degree in the School of Education. LDT comprises three concentrations: Learning Experience Design (LXD, currently operating), AI Leadership in Education (AILE, launching August 2026), and Learning Engineering (LENS, launching August 2026). LENS graduates demonstrate competency across five domains; the CLOs below are the assessable form.

THE FIVE CLOS

CLO-1 · Systems Thinking and Analysis. Decompose system performance requirements into measurable human capability requirements. Apply systems engineering lifecycle models. Analyze interdependencies between human and system performance to identify gaps and predict operational impact at scale.

CLO-2 · Learning Engineering Design and Implementation. Design capability development interventions through iterative engineering cycles. Evaluate evidence-based design strategies for transfer to high-consequence settings. Construct and communicate implementation plans.

CLO-3 · Human-System Collaboration and Adaptive Systems. Design human-system partnerships that make people more capable while preserving human agency. Specify delegation with revocation. Measure collaboration as a unit of analysis.

CLO-4 · Data, Measurement, and Evaluation. Design ethical instrumentation. Construct decision-grade evidence under irreducible uncertainty. Apply gap attribution. Demonstrate fairness beyond omission.

CLO-5 · Context and Domain Fluency. Integrate capability interventions into regulatory, organizational, cultural, and technical realities. Synthesize across boundaries. Anticipate adoption and sustainment barriers, including cross-regime / platform-dependency governance.

COURSE-TO-CLO COVERAGE

Course	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
LDT F1 (Learning Sciences Studio)		✗			
LDT F2 (Critical Perspectives)					✗
LEN 1 (Principles of LE for Complex Systems)	✗	✗			
LEN 2 (Human-AI Teaming and Adaptive Learning)			✗		
LEN 3 (Learning Engineering Systems)	✗				✗
LEN 4 (Evidence, Analytics, Measurement)				✗	
LEN 6 (Applied Problem Solving in LE)	✗	✗			✗
LEN 10 (Capstone)	✗	✗	✗	✗	✗
LEN 5 (Human Capability Analysis, elec.)	○				○
LEN 7 (Bias, Risk, Governance, elec.)					○

LEN 8 (Knowledge Transfer & Org Learning, elec.)		○			○
LEN 9 (Computational and AI Methods, elec.)			○	○	

✗ = required course; ○ = elective. Foundations + concentration + methods + capstone = 8 required courses (24 credits); 2 electives complete the 30-credit degree.

SECTION

Crosswalks

INDUCED 8 (CASEBOOK SCAFFOLD) → CANONICAL 5 (PROGRAM OF RECORD)

The casebook induces an 8-cluster analytic scaffold from its 194 cases (full account: `competencies.md`). Those eight cluster cleanly into the canonical five LENS competencies — except for D2 Iterative Development, which is the iteration **method** threaded through the cases rather than its own cluster. The gap is what the editor-commissioned first-person Practice Flywheel accounts will close in the next edition.

Induced cluster (casebook)	Canonical D	Notes
1 Capability requirements under operational reality	D1	Engineered vs. stated requirements; sustainment-as-requirement.
2 Evidence architecture the institution cannot deceive itself with	D4	Measurement gaming; post-deployment surveillance.
3 Interface and role design at the human-automation boundary	D3	Cues, alerts, mode/state transparency, recoverability.
4 Pairing mechanism with authorization	D3	Frontline halt authority; non-punitive reporting; authority gradient.
5 Governance architecture for deployment	D5	Consent, oversight, accountability as design artifacts.
6 Cross-organization and cross-time knowledge transfer	D5	Institution-building after catastrophe; tacit-knowledge survival.
7 Capability under system change and aging assumptions	D5	Legacy assets; multi-layer drift; transitions.
8 Equity and construct definition as design commitments	D4	Construct choice; demographic stratification; fairness.
(none specific)	D2	The iteration **method** — threaded across cases, not its own cluster.

THE CASEBOOK'S THREE ANCHORS PER CASE

Every v2 case carries three anchor codes in its header and metadata:

`induced-anchor` — e.g. `"2.4"` — the casebook's induced 8 / 32 sub-cluster the case evidences (analytic scaffold).

`lens-anchor` — e.g. `"D3/PT5"` — the canonical LENS five domain + problem type (1 – 6) the case is **primary** for.

`clo-anchor` — e.g. `"CLO-3"` — the assessable CLO the case serves as a worked example for (course mapping anchor).

SECTION

Casebook indexes

The same indexes that ship with **Capability Matters: A Casebook**, in v2.1 form. They are queried from the casebook source so they stay in sync as cases are added or revised.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset's standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE — **Case 5** — **Operation Eagle Claw (1980).** Cross-organizational capability that existed on no service's deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.

STANDOUT SUCCESS — **Case 18** — **U.S. Nuclear Navy / Rickover Training Model (1954–).** The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

1	USS Fitzgerald & USS John S. McCain 2017	T·K·N
2	Marine Corps Training in the INDOPACOM AOR ongoing	T·K
3	F-35 Sustainment & Maintainer Shortage ongoing	T·K·D
4	Military Fratricide — Desert Storm to Afghanistan 1991 – 2014	T·H·K
5	Operation Eagle Claw 1980	T·K
8	Patriot Missile / Dhahran 1991	D·H·K
17	GAO Weapon-System Sustainment Reviews — Operating Costs without Decision-Grade Data 2022 (GAO-22-104533)	G·K
18	U.S. Nuclear Navy / Rickover Training Model 1954 – present	T·K·N
21	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard 2020 revision (series since 1968)	G·K
33	GIFT and the Adoption Gap 2012 – present	K·G·N

AVIATION

STANDOUT FAILURE — Case 7

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS — Case 70

Crew Resource Management & CAST (1981–). First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

83	Atlas Air Flight 3591 2019	T·K
88	F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap 2008 – 2012	H·K·N
111	Aviation Safety Reporting System (ASRS) 1976 – present	T·K·N
118	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation 1996 – 2002	H·K·G
119	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson 1981 – 2008 (TCAS II Version 7.1)	H·K·G
164	NASA Saturn V Documentation 1972 – present	K
175	Singapore Airlines Safety Transformation 1980s – present	T·N

HEALTHCARE

STANDOUT FAILURE — **Case 61** — **EHR / CPOE Implementation (2005–).** Roughly $30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT SUCCESS — **Case 109** — **WHO Surgical Safety Checklist (2008–).** One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

20	California Nurse Staffing Ratios — Legislating a Capability Requirement 2004 – 2010	G·K
32	ACGME 80-Hour Resident Duty-Hour Reform 2003–2017	T·K·N
34	Implementation Science in Healthcare — The 17-Year Gap ongoing	K·G·N
59	Therac-25 1985 – 1987	H·D·G
61	EHR / CPOE Implementation 2005 – present	H·D·G
71	Keystone ICU / Pronovost Checklist 2004 – present	T·N
74	TREWS / Sepsis Watch 2018 – 2022	H·K·G
75	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff 2024 – 2025	H·K·N
79	ChatGPT in Healthcare — Hallucination Cases 2023 – present	H·D
86	Mid Staffordshire NHS Foundation Trust 2005 – 2009	G·N·K
93	Epic Sepsis Model 2017 – 2021	D·G·N
102	Medical Errors as Systemic Failure 1999 – present	T·H·N·K·G

EDUCATION AT SCALE

STANDOUT FAILURE — Case 149

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS — Case 110

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

171	xAPI / Total Learning Architecture — Interoperability Gap	ongoing	K·G
194	The Discipline We Build Next	ongoing	T·K·N·G·H·D

ENERGY

STANDOUT FAILURE	Case 69	**Three Mile Island (1979).** Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.
STANDOUT SUCCESS	Case 172	**INPO and the Nuclear Academy (1979–).** The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

11	Sago Mine Disaster	2006	N·T·K
60	Northeast Blackout	2003	H·K
69	Three Mile Island	1979	T·H
85	Deepwater Horizon	2010	T·N·K
87	Upper Big Branch Mine Explosion	2010	N·G·K
137	Texas City BP Refinery Explosion	2005	N·T·K·G
142	Davis-Besse Reactor Head Corrosion	2002	N·K·G
143	Fukushima Daiichi	2011	N·G·K
154	Camp Fire / PG&E	2018	G·N·K
172	INPO and the Nuclear Academy	1979 – present	T·K·G

GOVERNMENT

STANDOUT FAILURE	Case 162	**Australia Robodebt (2016–20).** An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found "venality, incompetence and cowardice"; the scheme was linked to deaths, including suicide.
NOTE	Case 111	No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

14	Healthcare.gov Launch	2013	K·T·G
101	VA Wait-Time Scandal	2014	G·K·N
103	LIBOR Manipulation	2003 – 2012	G·N
131	Predictive Policing — PredPol	2011 – present	G·H·D
166	Bernard Madoff / SEC Failures	1992 – 2008	G·K·N

INDUSTRIAL

STANDOUT FAILURE — **Case 147**

Bhopal (1984). Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.

STANDOUT SUCCESS — **Case 73**

Toyota Production System / Andon Cord (1950s–). Any operator can stop the line. The smallest authority-architecture intervention in the dataset, and the most durably copied.

30	Kodak Digital Camera Stagnation	1975–2012	D·K·N
73	Toyota Production System / Andon Cord	1950s – present	N·G
84	Takata Airbag Inflators	2008 – 2023	D·G
92	Volkswagen Dieselgate	2015	D·G
139	GM Ignition Switch	2002 – 2014	D·G
147	Bhopal	1984	T·K·N·G
148	Grenfell Tower	2017	G·T·K·N
153	Hyatt Regency Walkway Collapse	1981	D·G

TECHNOLOGY

STANDOUT FAILURE — **Case 66**

UK Post Office Horizon Scandal (1999–2015). Institutional refusal to believe sub-postmasters' reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest-running operator-feedback failure in the dataset.

NOTE — **Case 132**

The technology dataset's success story is filed under Healthcare: AlphaFold (Case 132), the contemporary worked example of computational capability-engineering done as a discipline.

AI IN EDUCATION

K-12 EDUCATION

K-12 MATHEMATICS

K-12 SCIENCE

AIR TRAFFIC MANAGEMENT

ANESTHESIOLOGY

AUTONOMOUS SYSTEMS

STANDOUT FAILURE — **Case 62** — **Uber ATG / Tempe Fatality (2018).** The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.

NOTE — **Case 194** — No paired-intervention success in the autonomous-systems dataset. The capability engineering this category needs is still being built — see Case 194 (**The Discipline We Build Next**) for the open question this volume closes on.

62	Uber ATG / Tempe Fatality 2018	T·N·G·H
78	Tesla Autopilot — Recurring Fatalities 2016 – present	T·N·G·H
192	Cruise Robotaxi — Pedestrian Drag 2023	G·D·H

AUTONOMOUS VEHICLES

156	Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment 2023	G·K·N
183	Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact 2023 / 2025	G·K·N
184	CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver 2020 – 2024	G·K·D

AVIATION INFRASTRUCTURE

23	FAA NextGen and the ADS-B Out Transition 2003 – 2020	G·D·K

AVIATION SAFETY

22	FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule 1988 – 2010s	G·D·K
180	FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike 1992 – 2000s	G·K·N

CLINICAL CARE

43	I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer 2014	H·K·N
120	Bar-Code Medication Administration — Cue at the Point of Care 2010	H·K·D

CLINICAL MEDICINE

CLINICAL/TRANSLATIONAL RESEARCH

CORPORATE L&D

CRIMINAL JUSTICE

DATA GOVERNANCE

DIGITAL IDENTITY

E-GOVERNMENT

ED-TECH

EDUCATION (NORWAY)

EDUCATION AT SCALE

FINANCE

FINANCIAL SERVICES

GENERAL AVIATION

GLOBAL HEALTH

GOVERNMENT

GOVERNMENT/WELFARE (NETHERLANDS)

HEALTH PROFESSIONS EDUCATION

HEALTHCARE/ONCOLOGY

HIGHER EDUCATION

HIGHER EDUCATION (AFRICA)

HIGHER EDUCATION (BRAZIL)

HIGHER EDUCATION (LATIN AMERICA)

HIGHER ED (UK)

HIGHER-ED ANALYTICS

HOSPITAL PHARMACY

IMPLEMENTATION SCIENCE

LEARNING ANALYTICS

MEDICAL DEVICES

MEDICAL EDUCATION

MEDICAL-DEVICE REGULATION

MULTIMODAL LEARNING ANALYTICS

NAVAL ENGINEERING

NEUROSCIENCE

SPACE / AEROSPACE

STANDOUT FAILURE	Case 140	**Challenger & Columbia (1986 / 2003).** Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine.
NOTE	Case 18	No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 18) for the safety-engineering counterpart that aerospace did not build.

9	Mars Climate Orbiter — Unit Mismatch 1999	D·K
15	Ariane 5 Flight 501 1996	D·K·H
65	NASA EVA-23 — Water Intrusion Inside a Spacesuit 2013	H·N·D
140	Challenger & Columbia 1986 / 2003	N·K·G
165	Boeing Starliner 2019 – 2024	K·D

SURGERY

121	Surgical Skill Video Peer-Rating Predicts Complications 2013	H·D·K
122	Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units 2009 – 2016	T·K·H

TECHNOLOGY

29	Amazon Hiring AI — Trained Bias, Deprecated 2018 2014 – 2018	D·K·N

TUTORING

77	Hybrid Human-AI Tutoring — Augmentation, Not Delegation 2024	T·K·D

WORKFORCE DEV

39	CIRCUIT — A Scalable, Equity-Centered Research Workforce Model 2017 – 2023 (six cycles)	T·K
185	Singapore SkillsFuture — National Workforce Capability at Scale 2015 – present	G·K·D

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

LENS COURSE INDEX

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

LEN 1

34 cases

LEN 2

53 cases

LEN 3

51 cases

LEN 4

65 cases

LEN 5

85 cases

LEN 6

17 cases

LEN 7

106 cases

LEN 8

92 cases

LEN 9

28 cases

LEN 10

19 cases

APPENDIX · CLOS AND COURSE COVERAGE

The competencies, the CLOs, and the courses.

COMPETENCY → CONCENTRATION LEARNING OUTCOME

The five capabilities the cases turn on are named, in the curriculum, as five concentration learning outcomes (CLOs):

Systems Analysis	CLO-1 · Systems Thinking and Analysis
Iterative Development	CLO-2 · Learning Engineering Design and Implementation
Test and Evaluation	CLO-3 · Data, Measurement, and Evaluation
Navigating Sociotechnical Constraints	CLO-4 · Context and Domain Fluency
Human-System Collaboration	CLO-3 · Human-System Collaboration and Adaptive Systems

THE LEN COURSES · CASES PER COURSE

Each case names the LEN courses it serves as a worked example for, and most cases serve several. The counts below are therefore not a partition of the 202 cases — they sum to well more than 202, because a single case typically appears under three or four courses.

LEN 1	Principles of learning engineering	36
LEN 2	Human-AI teaming	54
LEN 3	Systems integration	49
LEN 4	Evidence and measurement	67
LEN 5	Capability analysis	86

LEN 6	Applied problem-solving	18
LEN 7	Bias and governance	114
LEN 8	Knowledge transfer	98
LEN 9	Computational methods	30
LEN 10	Capstone studio	19

WHAT THE DISTRIBUTION SHOWS

The distribution is itself diagnostic. The strongly covered topics are the ones the existing case literature most readily supports: bias and governance (LEN 7) and knowledge transfer (LEN 8) lead, with capability analysis (LEN 5) and evidence and measurement (LEN 4) close behind. Across 202 cases the corpus is saturated with examples of what goes wrong when ethics, governance, evidence, and institutional learning are not engineered — and, in the success cases, what it takes to engineer them.

The thinner coverage is equally informative. Computational methods (LEN 9) has the failure modes of algorithmic systems well represented but the computational tools for capability instrumentation under-represented. The capstone (LEN 10) maps to fewer cases than the number of studio projects each cohort undertakes. Systems integration (LEN 3) and applied problem-solving (LEN 6) required focused tagging to reach defensible coverage, surfacing that operational systems-engineering content is under-represented in the published case literature relative to its curricular centrality, and that stakeholder-analysis and problem-framing have not historically been packaged as case material by other disciplines. These are pedagogical gaps the LENS studio sequence is organized to close — not by adding more failure cases, but by having students produce the missing artifacts in real operational settings and entering them into the dataset for the next cohort.

APPENDIX · REFERENCES BY CASE

References, case by case.

Every reference cited in every case, organised by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds an explicit **Retrieved from:** line pointing to the canonical access path — a publisher URL, a DOI, an agency landing page, or a stable archival locator. Where that line is left blank, the source is under verification; the running status of that work is in `casebook/verification-log.md`. Future editions will fill in remaining locators as the per-case review closes them out.

> **Style.** References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

CASE 1 USS Fitzgerald & USS John S. McCain 2017

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2017 – 2023 (six cycles)

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CASE 50 The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension

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2011 – present

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CASE 139 GM Ignition Switch

2002 – 2014

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CASE 140 Challenger & Columbia

1986 / 2003

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CASE 141 V-22 Osprey

1991 – present

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CASE 142 Davis-Besse Reactor Head Corrosion

2002

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